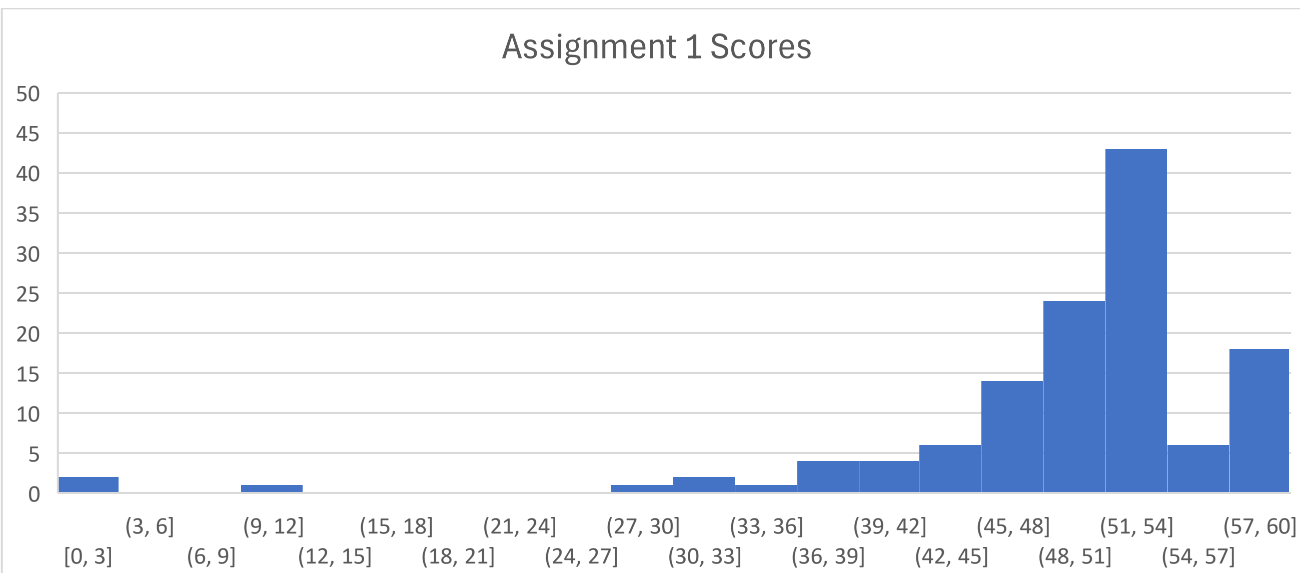
Assignment 1 SOLUTION: Chapters 1-5

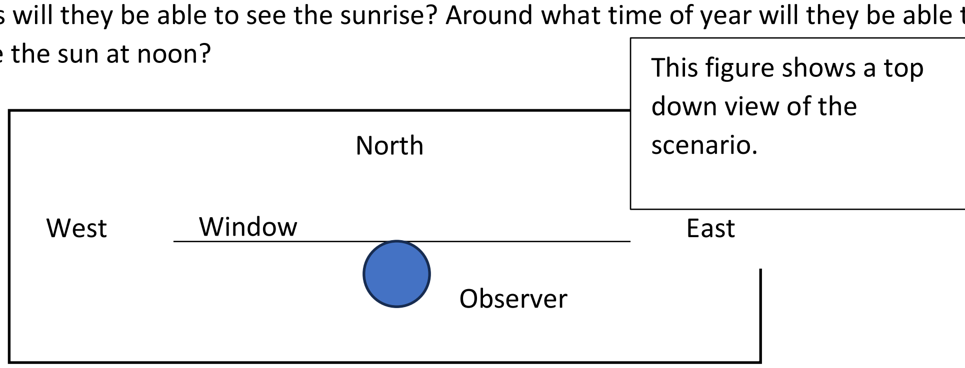
You may find your score from 60 marks in the Moodle Gradebook along with comments for the different questions explaining why you were marked as you were. Together with this solution, it is hoped that you can figure out how to improve your score if needed. Key concepts that I looked for in your solutions are highlighted in bold text below.

Mean Score: 49.9/60
Median Score: 52/60

52-60 ~ A
43-51 ~ B
37-42 ~ C
31-36 ~ D
Less than 30 ~ F



1. A person living in a Hong Kong high-rise apartment has a large window with an unobstructed view allowing them to see all locations of the sky that are to the North of East and West (All altitudes from $0°$ to $90°$; all azimuths from $270°$ to $360°$ and from $0°$ to $90°$). During which seasons will they be able to see the sunrise? Around what time of year will they be able to glimpse the sun at noon?



In the annual motion of the sun, the **sunrise and sunset position are directly East and West, respectively on the Vernal Equinox.** These positions shift **North until the day of the Summer Solstice and then return South. The sunrise and sunset are respectively East and West again on the Autumnal Equinox**. This means that the sunrise and sunset locations can be seen from the Vernal Equinox until the following Autumnal Equinox.

This means we can see the sunrise and sunset from a North facing window in the seasons of Spring and Summer, but not Autumn and Winter.

The daily path of the sun is inclined toward the south in northern latitudes. However, **Hong Kong is located between the Equator and the Tropic of Cancer which means there are two days of the year where the sun passes through Zenith**. Between those days, it would be possible to glimpse the sun at noon.
10 marks

I was confused by some explanations in the first part. Some students gave a response that felt like circular logic. Those responses may look like, they will be able to see the sunrise from Vernal Equinox to Autumnal equinox because those are the times that the sun rises to the North of East. This is true, but it doesn't explain why the sun rises to the North of East during that half of the year so it ends up sounds like, "They can see it at those times because those are the times it can be seen." Better explanation describes the annual motion of the sun.

2. Chloe works in Tokyo. Every day, around noon, she walks west from her office to her favorite café for lunch. She enjoys the shade of the buildings that line either side of the street. She notices that during some part of the year, the shadows of the buildings are long enough as to provide shade on both sides of the street, while in other times of the year one side of the street is in the sunlight.
   a. Which buildings are providing the shade, the ones on the north or south side of the street?

      Japan is in a northern temperate region**. In such regions the shadow of the noon sun always points north**. Thus we can conclude that the buildings on the south side of the street are providing the shade.
      5 marks

   b. During which time of year are the shadows longest?

      In northern temperate regions, the noon altitude of the sun is minimum on winter solstice and maximum on summer solstice. This means the we expect noon shadows to be longest around the winter solstice.
      5 marks

      Almost everyone got full marks on this question.

3. Over the course of many nights, a northern observer locates the same star and watches its daily motion. She observes that no matter what time of night or what time of year she always can spot this star in the sky and she never witnesses it rising above or setting below the horizon. In which direction (North, South, East, West) does she face to watch this star?

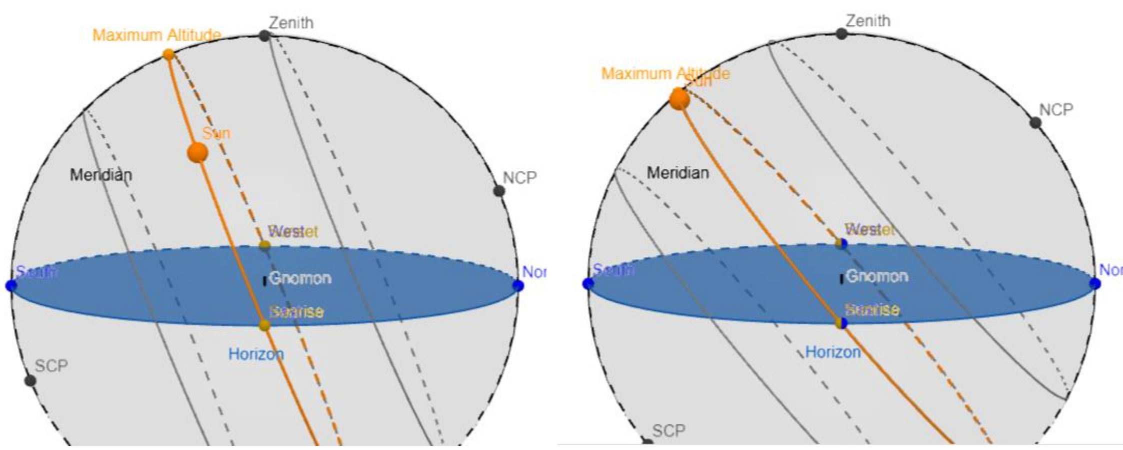
   If we look North, we see stars making counter clockwise circles around the North celestial pole. **For these stars nearest to the North celestial pole, they may complete a full circular path above the horizon.** It is these circumpolar stars that she watches. So she must be facing north.

   Only stars in this direction will could never observed but never observed to be rising or setting.

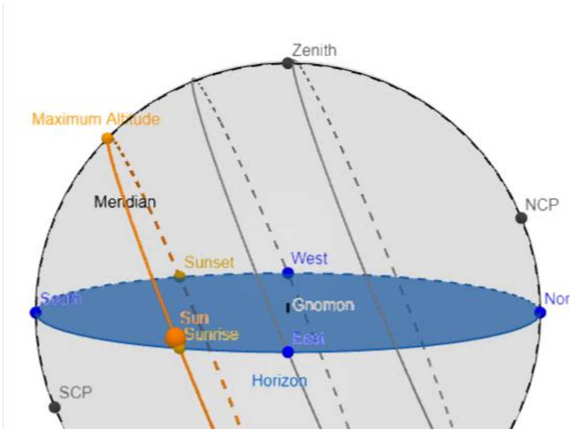
   10 marks

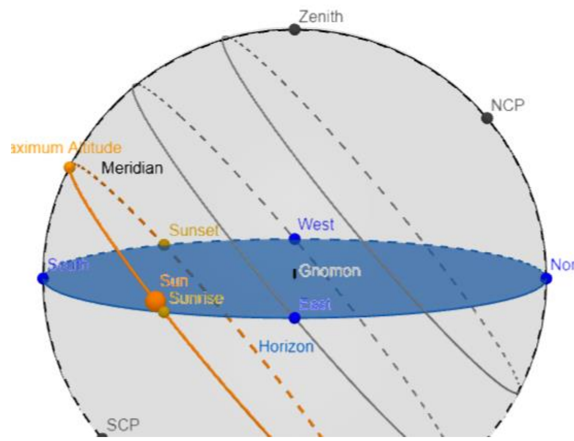
   This question was also well-performed. Some students identified the star as Polaris or the Pole Star or the North star. These are all names for the same star and a description of that star fits the description of what she observes, but there are many stars that circumpolar and all would fit this description. I tended to make a comment to this effect but not deduct marks.

4. Consider observations of the sun's daily and annual motions. Compare and contrast observations made in Hong Kong and Beijing.

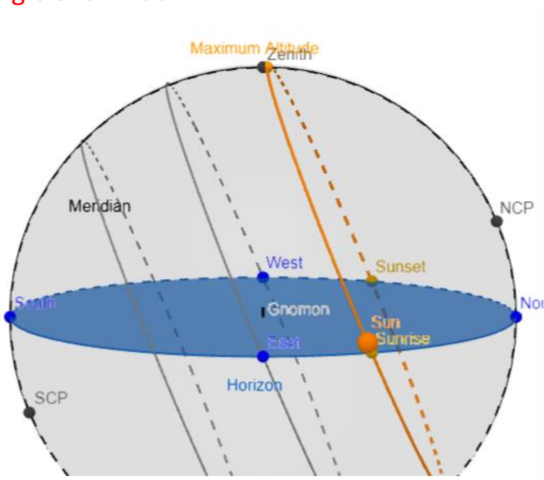


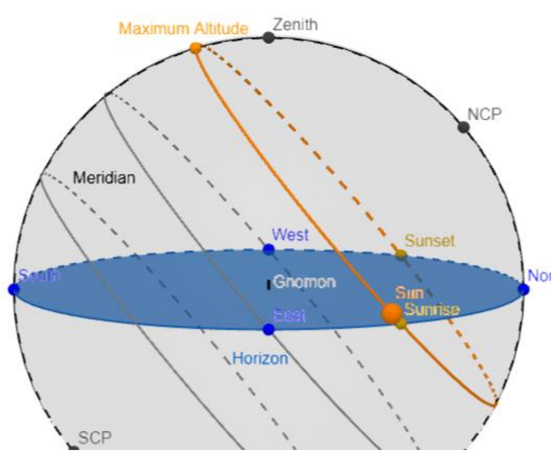
   Here we can compare the daily paths on Vernal or Autumnal Equinox. The Maximum Altitude or Noon Altitude is higher in HK than in BJ because of the smaller inclination in BJ. However, in both locations, exactly ½ of the daily path is above the horizon so the length of daylight on these days should be the same in the two locations. In both locations, the sunrises exactly east, so the azimuth of sunrise is exactly the same.





Next, we were asked to compare the daily paths between the two cities on Winter Solstice. Like on the Autumnal Equinox, the noon altitude is higher in HK, again due to the inclination being larger in HK. We can see that the sun rise location in BJ is further South in HK. This is a larger angle of azimuth.





Finally we are asked to compare the length of daylight. For the Vernal equinox, this is discussed in the first figure. For the summer solstice paths seen here, we see that only a small portion of the sun's path is below the horizon in BJ, while this is less extreme in HK. We expect a longer daylight in BJ.

a. Compare the noon altitude of the sun on the Autumnal Equinox. Repeat this comparison for the Winter Solstice.

The noon altitudes of the sun on the Autumnal Equinox and Winter Solstice will both be larger in Hong Kong than in Beijing. **Because of its higher latitude, the daily path of the sun is tilted further to the south in BJ than in HK.**
6 marks

b. Compare the azimuth of sunrise on the Winter Solstice. Repeat this comparison for the Vernal Equinox.

The shifting of sunrise position north and south in the annual motion is more extreme in more Northern latitudes. The winter solstice sunrise is further South or closer to 180 degrees azimuth in Beijing than in Hong Kong.

On the Vernal Equinox, the sun is expected to rise exactly East regardless of the observer's location.

If you used the gnomon applet to answer this, take care to observe the point labeled 'sunrise' to get a more insightful and accurate response.

6 marks

c. Compare the length of daylight on the Vernal Equinox. Repeat this comparison for the Summer Solstice.

The length of daylight should be the same in the two locations on the Vernal Equinox, equal day and night. We are halfway between the longest day, summer solstice and the shortest day, winter solstice. **However, the changes in length of daylight are more extreme at more northern latitudes.** Thus the length of the longest day, **Summer Solstice, should be longer in Beijing than in Hong Kong**.

6 marks

I worried about explanations in this question a fair amount. Some students gave explanations in part b that the sunrise location on Winter Solstice (WS) for BJ was further south because of its more northern latitude but then the sunrise location for BJ and HK were the same on the Vernal Equinox (VE), but gave no indication why latitude matters on WS but not VE. This means that latitude alone is not enough to explain the difference on WS. We must also add that we expect more extreme shifts to sunrise locations in more Northern latitudes.

5. In Lecture 3, we learned of several observations that led to ancient Greek astronomers concluding that the Earth is round. If an astronomer only observed the sky from a single location, what evidence, if any, could lead them to adopt a round Earth hypothesis?

An astronomer living close to the sea would wonder why you could see an approaching ship further away from the top of a tall mountain and why the hull of a departing ship disappeared before its mast. The Round Earth hypothesis would be a possible explanation for such an astronomer.

An astronomer that observed **a lunar eclipse** may deduce that the dark region the moon passes through was the Earth's shadow, indicating that the Earth has a round shape. The Round Earth hypothesis would be a possible explanation for such an astronomer.

12 marks possible

Full marks required giving both examples while it was very common to only list on. Any other incorrect examples also reduced marks.

Many of the observations that led to the Round Earth hypothesis depend on observing the path of the Sun and Stars at different locations to see how the Sun gives a different show for different people as one travels North and South. For example, the inclination of the sun's path changing with latitude, or the altitude of the North Celestial Pole**. Without traveling a significant distance North and South, it would be difficult to come to the conclusion of a Round Earth Hypothesis.**

Historically, it was the traveling North and South that caused ancient astronomers to conclude the Earth is Round. This means that perhaps other explanations explained the observations of sea-faring ships and the lunar eclipse.